BRUCE: Analytical exoplanet transit modelling and photometric analysis in Python and C

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Summary

A planet crossing its host star causes a small decrease in observed brightness. The shape and repetition of these transits constrain its relative size and orbit. Interpreting observations requires physical models, data preparation and parameter estimation. Repeated model evaluations become computationally demanding during fitting and searches of large datasets.

BRUCE is a Python and C package for analysing transiting exoplanets and related stellar systems. Its C modelling core is adapted from the Fortran-based `ellc` binary-star model (Maxted, 2016). It combines analytical transit and radial-velocity models with photometric processing, transit-template searches and parameter-estimation utilities. It also provides tools for retrieving observations from the Transiting Exoplanet Survey Satellite (TESS), investigating ambiguous periods and visualising orbital configurations. Its numerical routines are independently accessible. Source code and documentation are available in the BRUCE repository.

Statement of need

TESS (Ricker et al., 2015) has identified thousands of exoplanet candidates and produced extensive archives of light curves and full-frame images (Guerrero et al., 2021). Searching these data and fitting the resulting events require repeated model evaluations across many stars, epochs and trial parameters. The Next Generation Transit Survey (NGTS) (Wheatley et al., 2018) provides high-cadence survey photometry and extended monitoring of many long-period planet candidates from the TESS mission. Recovery campaigns (Gill et al., 2020b; Gill et al., 2020a; Gill et al., 2020c) and subsequent work (Gill et al., 2024) exemplify the large, often discontinuous datasets that must be searched for sparse transit signals. Efficient models are therefore needed both to scan these observations and to fit events jointly with TESS data. PLATO's planned long-baseline observations of large numbers of stars (Rauer et al., 2025) will extend the need for computationally efficient detection and characterisation tools.

BRUCE combines fast modelling with data retrieval, normalisation, binning, template searches and period-alias assessment. A fitted transit can become a search template, and the recovered events can constrain a planet's orbital period. This package provides a toolkit to find and fit exoplanet systems in photometric data and allows users to develop custom fitting routines as required.

State of the field

Several established packages address parts of this problem. `batman` provides transit calculations for a range of limb-darkening laws (Kreidberg, 2015). `ellc` models detached eclipsing binaries and transiting planets with a broader treatment of stellar geometry and binary effects (Maxted, 2016). PyTransit provides efficient transit models and interfaces designed for demanding photometric analyses (Parviainen, 2015), including a documented power-2 implementation. Lightkurve provides an extensive environment for accessing and analysing Kepler and TESS observations (Lightkurve Collaboration, 2018).

BRUCE builds on these foundations. A dedicated C layer combines model evaluation, likelihood reduction and transit-search loops; a common Python interface connects these operations to data retrieval and period-alias analysis. This integrated workflow motivates a separate package rather than extending a single transit-model library. The package focuses on speed and provides a single environment in which exoplanet researchers can find transit signals, fit planetary systems and interpret results.

Software design

The binary-star modelling core is adapted from `ellc` (Maxted, 2016), with an analytical eccentric-anomaly solver based on Markley (1995): a cubic approximation followed by a fifth-order correction solves the elliptic Kepler equation. The resulting true anomaly determines Keplerian radial velocities and the projected separation used for photometric transits. Transit evaluation focuses on the fast analytical `qpower2` approximation (Maxted & Gill, 2019), using the power-2 limb-darkening law. Avoiding repeated numerical integration over the stellar disc makes this approximation efficient for fitting exoplanet transits but comes at the cost of accuracy as the radius ratio increases (see (Maxted & Gill, 2019)). Figure 1 compares the analytical transit model with independent numerical integration for a typical exoplanet system.

Python supplies array-based interfaces, data handling and fitting orchestration, while C with OpenMP handles repeated calculations. NumPy arrays connect these layers through the NumPy C API. OpenMP distributes sample-wise calculations across CPU threads, further accelerating fitting and minimisation routines. A combined evaluation mode accumulates a likelihood statistic without returning the full model array, although its performance advantage depends on workload and memory costs. Numerical cadence integration requires odd subsample counts of at least three, with a default of 11, centred across each exposure. Keplerian radial velocities support primary and double-lined systems, and sampling helpers interface with `emcee` (Foreman-Mackey et al., 2013) and SciPy-based minimisation routines (Virtanen et al., 2020).

The higher-level modules support TESS data retrieval, photometric time-series preparation, transit fitting, template matching and period-alias assessment. Detrending remains an explicit modelling choice: this package provides filter routines that can remove stellar variability but also distort a transit if not used appropriately. Plotting, sampling and minimisation helpers and orbit animations support interpretation of fitted systems.

Use cases

Direct transit fitting. BRUCE provides fitting routines and sampling utilities that simplify preparing photometric data, fitting transit models and interpreting the results. Supplying photometry and uncertainties to `bruce.binarystar.lc` evaluates the model and accumulates the likelihood statistic in C, avoiding transfer of a model array at every optimiser call or sampler step.

Detecting single transits. Transit templates can be compared to photometric data over a range of trial epochs to search for transit events. Published examples of this method include NGTS-11 b (Gill et al., 2020c) and TOI-2447 b/NGTS-29 b (Gill et al., 2024). The latter uses a TESS-derived template to recover events in archival NGTS survey observations and subsequent monitoring. These methods apply to both ground-based and space-based photometry, including searches through TESS data for additional transits of long-period planetary systems. Such searches can be computationally expensive, but BRUCE provides fast template matching capabilities. Detection thresholds must be calibrated for each dataset and search.

Resolving duotransits. Two separated transits permit discrete aliases. BRUCE provides a period-resolution framework that organises transit events, generates candidate aliases and combines event fitting, observing coverage and searches for additional events to assess the possible periods. This framework simplifies the analysis of duotransit systems such as those identified in TESS samples (Hawthorn et al., 2024).

Model comparison

BRUCE is compared with `ellc` numerical integration using identical power-2 coefficients [Figure 1]. Coefficients follow Maxted (2018). No parameters are fitted. For this case, the maximum difference is 74.0 ppm; changing `ellc` from fine to very-fine grids changes the reference by at most 0.58 ppm. Source code and model arrays accompany the paper.

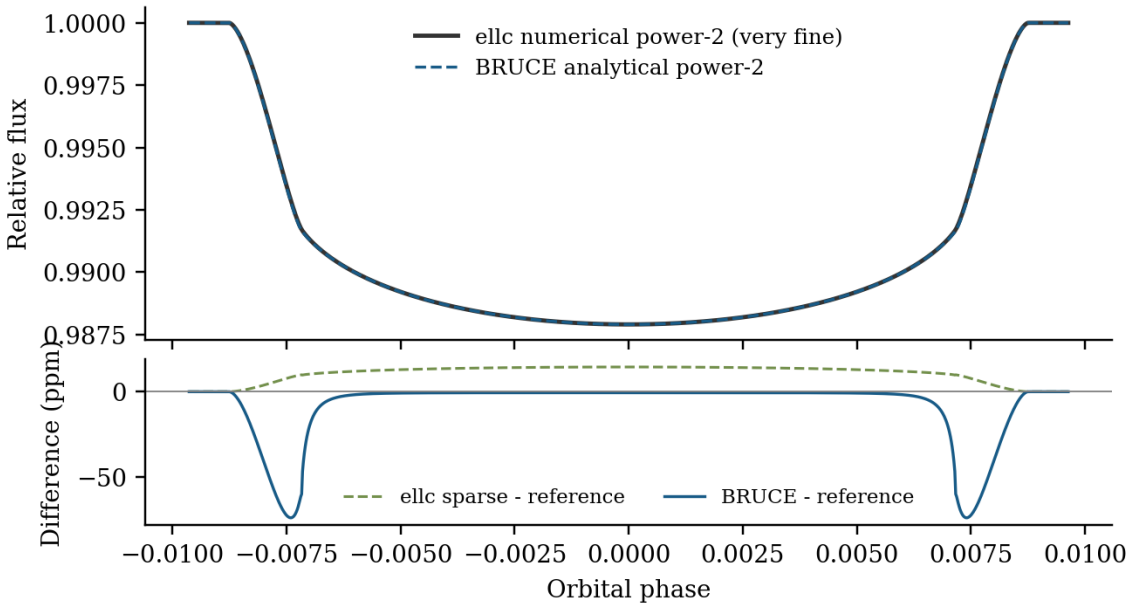


Figure 1. Analytical versus numerical power-2 transit models for spherical bodies with scaled stellar radius 0.05, radius ratio 0.1, zero impact parameter and eccentricity, and instantaneous sampling. Both use c=0.715452 and alpha=0.643777. Upper: BRUCE and very-fine ellc integration. Lower: differences from that reference, including sparse ellc integration for context. This matched-law comparison follows the layout of Maxted & Gill (2019), Figure 3.

Computational performance

BRUCE is a fast light-curve framework and its speed is demonstrated with a benchmark test that records model-only calls and direct C likelihood calls with `flux` and `flux_err` supplied to `bruce.binarystar.lc`. Figure 2a shows the latter, the relevant repeated evaluation for fitting. Contiguous double-precision arrays contain 100 to 100,000 samples; seven batches follow warm-up in separate one- and four-thread processes. The instantaneous circular case uses e=0, the instantaneous eccentric case e = 0.4, and the exposure-integrated case e=0 with 11 subsamples over 2 minutes. All use fixed uncertainties. Imports, input construction, I/O and optimiser execution are excluded; settings, raw timings, source hashes and the compiler log are retained.

On an Intel Core i7-9750H at a nominal 2.60 GHz, with six physical cores, Python 3.9.13, NumPy 1.26.4 and Apple Clang 21.0.0 with OpenMP, the 1,000-sample circular likelihood achieved approximately 7,500 evaluations per second with one thread and 28,000 with four. For 100,000 samples, the rates were 57 and 280 evaluations per second, a speed-up of 4.90. At that size, the one-thread eccentric and exposure-integrated rates were 25.6 and 11.0 evaluations per second. Rates describe complete model-and-likelihood evaluations, not independent posterior samples.

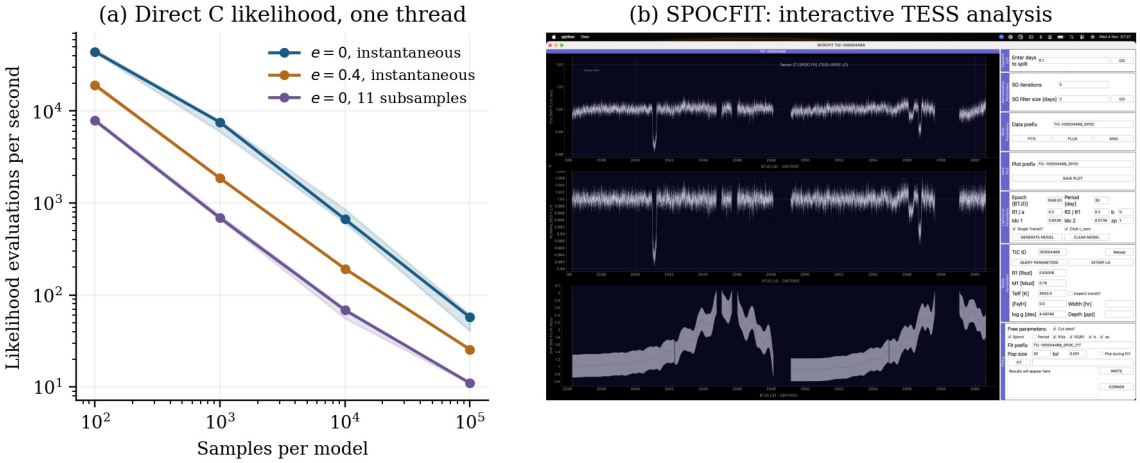


Figure 2. Numerical throughput and interactive TESS analysis. Left: reciprocal median call time with one thread; shading spans reciprocals of the upper and lower timing quartiles. Right: the SPOCFIT GUI displaying TIC 100504488 in TESS sector 27, with input and normalised photometry, sky background and controls for detrending and transit fitting. The throughput measurements are local BRUCE benchmarks; the screenshot illustrates the interactive workflow.

These tests are hardware dependent. Users should be aware of the`OMP_NUM_THREADS` environment variable to control the number of threads used when calculating a log-likelihood or generating a model. Users should set it to 1 when fitting data with multiprocessing (e.g. emcee).

Research impact statement

BRUCE grew from the need to search and fit large photometric datasets efficiently, particularly TESS and NGTS long-period candidates. The author's recovery programme includes TIC-238855958, TIC-231005575 and NGTS-11 b (Gill et al., 2020b; Gill et al., 2020a; Gill et al., 2020c), followed by TOI-2447 b/NGTS-29 b (Gill et al., 2024); the TESS duotransit catalogue of Hawthorn et al. (2024) further motivates tools for ambiguous periods and targeted monitoring. These studies motivate BRUCE's workflows; they do not all use the present release.

BRUCE also provides undergraduate and postgraduate students, as well as established researchers, with tools to find, fit and prepare planetary-system analyses for publication. The `spocfit` GUI (Figure 2b) provides an interactive route to inspect SPOC photometry, adjust normalisation, overlay transit models and export processed data. Its GUI tools offer early-career astronomers an accessible starting point for TESS data analysis: students can explore light curves and modelling choices interactively before developing scripted analyses. Reproducible tests, benchmarks and regression checks are provided.

AI usage disclosure

OpenAI Codex assisted with manuscript drafting, bibliography preparation, benchmark and figure scripts, documentation, and preceding numerical refactoring and regression-test development. The author subsequently edited the text substantially and checked that the text and figures reflect the intended content. The author used Codex assistance for the NumPy C API bindings. Automated checks included compilation and regression tests.

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